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Inventor(s)

De Giovanni; Pierfrancesco et al.

SYSTEMS AND METHODS FOR ENGINE VALVE COOLING

Abstract

A valve for a pressurizable enclosure includes a valve head provided at a first portion of the valve, a connecting portion extending between the first portion and a second portion of the valve, a chamber comprising a hollow portion, and a first body disposed within the chamber. The valve head is configured to selectively permit fluid flow into or out of the pressurizable enclosure. The hollow portion of the chamber is provided at least partially within the connecting portion and configured to contain a filler material. The first body is configured to be movable within the chamber during valve operation.

Inventors: De Giovanni; Pierfrancesco (Corsano, IT), Aimo Boot; Marco (Nole, IT), Lia; Luigi (Torino, IT)

Applicant: Eaton Intelligent Power Limited (Dublin 4, IE)

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Background/Summary

PRIORITY [0001] This application is a continuation under 35 U.S.C. § 365(c) of International Patent Application No. PCT/EP2023/025446, filed on 25 Oct. 2023, which claims the benefit under 35 U.S.C. § 119(e) of U.S. Provisional Patent Application No. 63/380,820, filed on 25 Oct. 2022, all of which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

[0002] The present disclosure relates generally to valves and, more particularly, to valves designed for improved cooling performance.

BACKGROUND

[0003] Engine valves can experience large thermal loads due to the flow of hot gases from engine cylinders. The large thermal loads experienced by valves may cause them to fail. There is a need to improve cooling of valves during engine operation to facilitate reduced operating temperatures, improved performance, and extended operating life of the valves.

SUMMARY

[0004] In particular embodiments, a valve for a pressurizable enclosure is disclosed, the valve including: a valve head provided at a first portion of the valve and configured to engage with the pressurizable enclosure to selectively permit fluid flow into or out of the pressurizable enclosure; a connecting portion extending between the first portion and a second portion of the valve provided opposite the first portion; a chamber including a hollow portion provided at least partially within the connecting portion and configured to contain a filler material; and a first body disposed within the chamber; wherein, during valve operation, the first body is configured to be movable within and relative to the chamber to facilitate reducing an operating temperature of the valve.

[0005] In particular embodiments, which may combine the features of some or all of the above embodiments, the filler material includes a fluid medium during valve operation, and wherein the first body is at least partially immersed in the fluid medium during valve operation. In particular embodiments, which may combine the features of some or all of the above embodiments, valve operation includes a periodic translation or a reciprocal motion of the valve relative to the pressurizable enclosure along a longitudinal axis.

[0006] In particular embodiments, which may combine the features of some or all of the above embodiments, the chamber includes an elongated portion oriented along a longitudinal axis of the valve, and wherein the first body is configured to reciprocate or translate within the elongated portion of the chamber during valve operation. In particular embodiments, which may combine the features of some or all of the above embodiments, the first body includes an elongated portion provided coaxially with the elongated portion of the chamber.

[0007] In particular embodiments, which may combine the features of some or all of the above embodiments, the valve further includes one or more second bodies disposed within the chamber, each second body configured to be movable within and relative to the chamber during valve operation.

[0008] In particular embodiments, which may combine the features of some or all of the above embodiments, the first body includes a first portion, a second portion provided opposite the first portion, and a third portion disposed between the first portion and the second portion. In particular embodiments, which may combine the features of some or all of the above embodiments, the third portion includes an elongated section oriented along a longitudinal axis of the valve. In particular embodiments, which may combine the features of some or all of the above embodiments, one or more of the first portion and the second portion are configured to taper along a direction away from the third portion. In particular embodiments, which may combine the features of some or all of the above embodiments, the first body includes a spherical portion or a hemispherical portion.

[0009] In particular embodiments, which may combine the features of some or all of the above embodiments, the valve is an intake valve or an exhaust valve of a combustion engine, and wherein a motion of the first body within and relative to the chamber during valve operation permits an increased rate of heat transfer through the connecting portion of the valve to facilitate reducing the operating temperature of the valve. In particular embodiments, which may combine the features of some or all of the above embodiments, the filler material includes sodium.

[0010] In particular embodiments, which may combine the features of some or all of the above embodiments, reducing an operating temperature of the valve includes reducing a mean temperature of the valve head during valve operation.

[0011] In particular embodiments, which may combine the features of some or all of the above embodiments, a method to facilitate reducing an operating temperature of a valve of a pressurizable enclosure is disclosed, the method including: providing a movable first body within a chamber of the valve, the chamber provided at least partially within a connecting portion extending between a valve head and a valve end of the valve, the chamber configured to contain a filler material, the filler material including a fluid medium during valve operation, the first body at least partially immersed in the fluid medium during valve operation; and operating the valve by periodically translating the valve relative to the pressurizable enclosure along a longitudinal axis such that the first body moves within and relative to the chamber to facilitate reducing an operating temperature of the valve.

[0012] In particular embodiments, which may combine the features of some or all of the above embodiments, the disclosed method further includes: providing one or more movable second bodies within the chamber, one or more of the second bodies at least partially immersed in the fluid medium during valve operation; and operating the valve by periodically translating the valve relative to the pressurizable enclosure along a longitudinal axis such that one or more of the second bodies move within and relative to the chamber to facilitate reducing an operating temperature of the valve.

[0013] In particular embodiments, which may combine the features of some or all of the above embodiments, a cylinder assembly for an engine is disclosed, the cylinder assembly including: a first pressurizable enclosure; a first piston configured to reciprocate within the first pressurizable enclosure; and one or more first valves configured to periodically operate based on a position of the first piston to selectively permit fluid flow into or out of the first pressurizable enclosure, wherein each first valve respectively includes: a valve head provided at a first portion of the first valve; a connecting portion extending between the first portion and a second portion of the first valve provided opposite the first portion; a chamber including a hollow portion provided at least partially within the connecting portion and configured to contain a filler material; and one or more movable bodies disposed within the chamber; wherein, during operation of the first valve, one or more of movable bodies are configured to translate or reciprocate within and relative to the chamber to facilitate reducing an operating temperature of the first valve.

[0014] In particular embodiments, which may combine the features of some or all of the above embodiments, the first pressurizable enclosure includes a cylinder of a combustion engine, and wherein at least one of the one or more first valves is an inlet valve or an exhaust valve. In particular embodiments, which may combine the features of some or all of the above embodiments, the filler material includes sodium.

[0015] In particular embodiments, which may combine the features of some or all of the above embodiments, each chamber corresponding to the one or more first valves respectively includes an elongated portion oriented along a longitudinal axis, wherein at least one movable body corresponding to each first valve includes a respective elongated portion, and wherein the elongated portion of the at least one movable body is provided coaxially with the elongated portion of the respective chamber.

[0016] In particular embodiments, which may combine the features of some or all of the above

embodiments, the cylinder assembly further includes one or more second pressurizable enclosures; one or more second pistons respectively configured to reciprocate within the one or more second pressurizable enclosures; and one or more second valves corresponding to each of the one or more second pressurizable enclosures, each second valve respectively configured to periodically operate based on a position of the corresponding second piston to selectively permit fluid flow into or out of the corresponding second pressurizable enclosure, wherein each second valve respectively includes: a valve head provided at a first portion of the second valve; a connecting portion extending between the first portion of the second valve and a second portion of the second valve provided opposite the first portion of the second valve; a chamber including a hollow portion provided at least partially within the connecting portion of the second valve and configured to contain a filler material; and one or more movable bodies disposed within the chamber of the second valve; wherein, during operation of the second valve, one or more of the movable bodies are configured to translate or reciprocate within and relative to the chamber of the second valve to facilitate reducing an operating temperature of the second valve.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] The following figures are included to illustrate certain aspects of the present disclosure to provide an understanding, and should not be viewed as exclusive embodiments. The subject matter disclosed contemplates and allows for considerable modifications, alterations, combinations, and/or equivalents in form and function, without departing from the scope of this disclosure.

[0018] FIG. 1 illustrates a perspective view of exemplary engine cylinders, according to particular embodiments.

[0019] FIG. 2A illustrates a cross-sectional view of one of the engine cylinders of FIG. 1, with the piston at top dead center, according to particular embodiments.

[0020] FIG. 2B illustrates a cross-sectional view of one of the engine cylinders of FIG. 1, with the piston at bottom dead center, according to particular embodiments.

[0021] FIG. 3 illustrates a cross-sectional view of an exemplary valve, according to particular embodiments.

[0022] FIG. 4 illustrates exemplary shapes of turbolators, according to particular embodiments.

[0023] FIG. 5 illustrates a graph comparing operating temperature profiles of exemplary valves, according to particular embodiments.

DETAILED DESCRIPTION

[0024] Illustrative embodiments of the present invention are described in detail herein. In the interest of clarity, not all features of an actual implementation may be described in this specification. It will of course be appreciated that in the development of any such actual embodiment, numerous implementation-specific decisions may be made to achieve the specific implementation goals, which may vary from one implementation to another. Moreover, it will be appreciated that such a development effort might be complex and time consuming but would nevertheless be a routine undertaking for those of ordinary skill in the art having the benefit of the present disclosure.

[0025] Throughout this disclosure, a reference numeral followed by an alphabetical character may refer to a specific instance of an element, and the reference numeral alone may refer to the element generically or collectively. Thus, as a non-limiting example (not shown in the drawings), widget “1a” may refer to an instance of a widget class, which may be referred to collectively as widgets “1” and any one of which may be referred to generically as a widget “1”. In the figures and the description, like numerals are intended to represent like elements. For clarity, not every element may be labeled or otherwise referenced in every figure. Additionally, not every element illustrated

by way of non-limiting example in a given figure need be present in every contemplated embodiment.

[0026] The terms “couple” or “couples,” as used herein, are intended to mean either an indirect or direct connection. Thus, if a first device couples to a second device, that connection may be through a direct connection, or through an indirect electrical connection or a shaft coupling via other devices and connections.

[0027] To facilitate a better understanding of the present disclosure, the following examples of certain embodiments are given. In no way should the following examples be read to limit, or define, the scope of the disclosure. Embodiments described below with respect to one implementation are not intended to be limiting.

[0028] The present disclosure provides for systems and methods for utilizing valves with features for enhanced cooling and/or lower operating temperatures in use. As will be discussed further, disclosed systems and methods may, in certain embodiments, modify rates of heat transfer in or through a valve. By way of example and not limitation, disclosed systems and methods may include one or more movable bodies disposed within a coolant or heat transfer medium provided within a valve. The disclosed valve features may be utilized in passenger cars, trucks, or any other suitable vehicle. While disclosed features may be used in engines, e.g., for engine cylinder valves, and may be described relative to engine cylinders to provide a better understanding, it should be appreciated that the disclosed features are contemplated in the context of valves for any suitable pressurizable containers or enclosures that may benefit from their modification, adaptation, and/or use.

[0029] In particular embodiments, an engine may comprise four, six, or eight cylinders, or any other number of cylinders. Other numbers of cylinders can be used without affecting the relevance or applicability of the present disclosure, but for discussion, particular numbers of cylinders may be illustrated in the figures (such as in FIG. 1 by way of example and not limitation). In particular embodiments, an engine may utilize a camshaft system to operate one or more valves, or it may use a camless design, or a hybrid system. By way of example and not limitation, the intake valves and exhaust valves may either couple to a cam system for actuation, a hydraulic rail, a latched rocker arm, other rocker arm, switching roller finger follower, lashed capsule, an electric actuator, a hydraulic actuator, and/or electro-hydraulic actuator, etc. Additional functionality, such as engine braking and hydraulic lash adjustment may be separately or additionally included. In particular embodiments, the cylinders of an engine may receive intake gases. By way of example and not limitation, intake gases may comprise combustible gases, such as air, and/or air mixed with fuel, and/or air mixed with exhaust (exhaust gas recirculation “EGR”), from an intake manifold. In particular embodiments, one or more intake manifold sensors may monitor the pressure, flow rate, oxygen content, exhaust content or other qualities of the intake fluid. In particular embodiments, the intake manifold may connect to intake ports in the engine block to provide intake fluid to the cylinders. In particular embodiments, an intake manifold may be provided at a vacuum relative to the local atmospheric pressure. In particular embodiments, the intake manifold may be boosted to a high pressure relative to the local atmospheric pressure.

[0030] In particular embodiments, fuel may be injected to individual cylinders via a suitable fuel distribution and/or injection system. By way of example and not limitation, a fuel distribution and/or injection system may comprise a fuel injection controller. In particular embodiments, a fuel injection controller may adjust the amount and timing of fuel injected into each cylinder, and may be configured to selectively shut off and resume fuel injection to each cylinder. In particular embodiments, a quantity of fuel injected for each cylinder of a multi-cylinder engine may be the same, or may be unique for each cylinder or particular groups of cylinders. By way of example and not limitation, one cylinder may receive more fuel than another, on a time-averaged basis, and another cylinder may receive no fuel at particular times, such as to conserve fuel during low power requirements, or for other operational reasons.

[0031] FIG. 1 illustrates a perspective view of example engine cylinders **100** that may employ the aspects of the present disclosure, according to particular embodiments. The engine block of a suitable engine utilizing the engine cylinders **100** is removed from the present example for clarity. In particular embodiments, operation of the engine cylinders **100** may be dependent on intake valves **102**, exhaust valves **104**, and pistons **106**. In particular embodiments, there may be a single intake valve **102**, and/or a single exhaust valve **104**, and a piston **106** per cylinder **100**. In other embodiments, each cylinder **100** may operate with a plurality of intake valves **102**, and/or a plurality of exhaust valves **104**. As shown in FIG. 1 by way of example and not limitation, piston **106** may be disposed within the cylinder **100**, and may be operable to translate within the cylinder **100**. In particular embodiments, each piston **106** may be coupled to a crankshaft **108** housed in the engine block. The crankshaft **108** may allow the engine to convert movement of one or more pistons **106** into rotational motion. In particular embodiments, crankshaft **108** may be coupled to transfer energy to one or more camshafts (such as **110**, **112**, by way of non-limiting example). By way of example and not limitation, crankshaft **108** may transfer energy to one or more camshafts by a torque transfer mechanism **114**, which may comprise one or more of gear sets, belts, and/or other transfer mechanisms. In particular embodiments, each intake valve **102** and/or each exhaust valve **104** may couple to a cam system, such as the camshafts **110**, **112**, respectively as non-limiting examples, for actuation and operation.

[0032] FIGS. 2A-2B illustrate one of the engine cylinders **100** at top dead center and bottom dead center, respectively according to particular embodiments. In particular embodiments, an engine may operate by compressing fluid (such as air, or a mixture of air and fuel, by way of non-limiting example) provided into a cylinder **100** through an intake valve **102** by compression using a piston **106**. In particular embodiments, once the intake fluid has moved from the intake manifold to the cylinder **100**, it may be referred to as a “charge,” and when the charge moves from the cylinder **100** to an exhaust manifold, it may be referred to as exhaust gas. In particular embodiments, to initiate a combustion phase of engine operation, fuel may be injected via a fuel injector (e.g., in a diesel engine), or a mixture of air and fuel may be ignited using a spark (e.g., in a gasoline engine). By way of example and not limitation, combustion forces due to combustion may force the piston **106** from top dead center (TDC) illustrated in FIG. 2A to bottom dead center (BDC) illustrated in FIG. 2B. In particular embodiments, torque may then be directed to the crankshaft **108** for output, such as to a driveshaft, and/or an affiliated flywheel. In particular embodiments using diesel engines, combustion initiation may be referred to as “compression ignition.” In particular embodiments, combustion initiation in a gasoline engine may use a spark plug to ignite the gasoline.

[0033] In particular embodiments, engine operation may encompass a 4-stroke operation. In particular embodiments, other operation modes such as 2-stroke, 6-stroke, and 8-stroke are also possible and contemplated herein. By way of example and not limitation, corresponding to a 4-stroke operation, during a first stroke (“stroke 1”) of a 4-stroke cycle, the piston **106** may move from TDC to BDC with one or more intake valves **102** may open, so as to fill the cylinder **100** with intake fluid. In particular embodiments, the intake fluid (such as air, and/or air mixed with fuel and/or exhaust) may flow from an intake manifold into each cylinder through intake ports **200** located in an engine block. In particular embodiments, with all valves closed, the piston **106** may then translate back to TDC (“stroke 2”) to compress the fluid drawn in during intake. In particular embodiments, fuel may be injected (e.g., in a diesel engine) and/or the fluid charge ignited by a spark (e.g., in a gasoline engine) to initiate combustion, thereby pushing the piston **106** to BDC (“stroke 3”). In particular embodiments, the piston **106** may then rise again to TDC with one or more exhaust valves **104** open so as to expel the exhaust out the exhaust valve **104** (“stroke 4”). In particular embodiments, the intake valve **102** may be open during stroke 1 and closed during strokes 2-4, wherein the intake valve **102** may be seated against the intake port **200**. In particular embodiments, the exhaust valve **104** may be open during stroke 4 and closed during strokes 1-3, wherein the exhaust valve **104** may be seated against an exhaust port **204**. In particular embodiments,

compression occurs on the second stroke, and combustion and/or power output occurs on the third stroke. In particular embodiments, 6-stroke and 8-stroke techniques may include additional aspects of compression and/or injection after the intake valve **102** has closed, and/or prior to the exhaust valve **104** opening. It will be appreciated that while the present disclosure describes 4-stroke combustion techniques in detail to provide an understanding, aspects of the disclosure are contemplated for applicability to other cycles and modes of operation. Separately or additionally, particular 4-stroke techniques may be applicable to 6-stroke or 8-stroke techniques.

[0034] In particular embodiments, the intake and exhaust valves, **102** and **104** respectively, may be actuated by valve actuators **202**. In particular embodiments, each valve **102**, **104** may be actuated by a dedicated valve actuator **202**, or actuation may be split or shared. By way of example and not limitation, a particular valve actuator may provide actuation of multiple valves, which may span multiple cylinders in particular embodiments, and/or cover groups comprising both intake and exhaust valve combinations. In particular embodiments, the valve actuator **202** may be any suitable component for operating engine valves, such as a mechanical, hydraulic, electric, and/or electric solenoid systems to control the intake and exhaust valves **102**, **104**. In particular embodiments, the valve actuators **202** for each cylinder **100** may be the same for all cylinders **100**, or may differ between the intake valves **102** and the exhaust valves **104** so that certain functionality may be only enabled on one or the other of those valves. In particular embodiments, when more than one intake valve **102** or more than one exhaust valve **104** are used per cylinder **100**, the valve actuators **202** may be the same or different for each of those valves. In particular embodiments, one or more valves, such as inlet valves **102** and/or exhaust valves **104**, may be operated by cams provided on camshafts to periodically engage with the valve.

[0035] In particular embodiments, exhaust gases may leave cylinders **100** through exhaust ports **204** in an engine block. In particular embodiments, exhaust ports **204** may communicate with an exhaust manifold. In particular embodiments, an exhaust manifold sensor may be configured to monitor the pressure, flow rate, oxygen content, nitrous or nitric oxide (NO_x) content, sulfur content, other pollution content or other qualities of the exhaust gas. In particular embodiments, the exhaust gas can power a turbine of turbocharger, which can be a variable geometry turbocharger (VGT) or other turbocharger. In particular embodiments, a turbocharger may be adjusted so as to control intake or exhaust flow rate or back pressure in the exhaust.

[0036] In particular embodiments, compressed gases, combusting gases, and/or combustion products may provide an environment proximal to one or more inlet valves **102** and/or exhaust valves **104** comprising high heat fluxes and loads, and/or elevated temperatures. By way of example and not limitation, exhaust gases leaving each cylinder **100** may impinge on each inlet valve **102**, and/or flow over and around each exhaust valve **104**. By way of example and not limitation, exhaust valves **104** may receive significant heat loads and heat fluxes, and/or experience high temperatures. In certain embodiments, the exhaust valves **104** may fail due to the increase in heat and elevated temperature. In particular embodiments, to increase cooling performance of the valves and/or to improve fatigue life of one or more valves, designs and features for enhancing cooling of the valves may be used. In particular embodiments, hollow valves may be used to improve cooling performance of the valves in operation. In particular embodiments, a heat transfer medium and/or filler medium may be used within one or more chambers located within a valve. In particular embodiments, a heat transfer medium and/or filler medium provided within a hollow chamber of a valve may perform and/or behave as a fluid medium at engine and valve operating conditions, such as due to operating conditions of elevated temperature relative to room temperature. In particular embodiments, one or more movable bodies may be further provided within one or more chambers or enclosures disposed within a valve.

[0037] In particular embodiments, one or more movable bodies disposed within the hollow valve may be at partially immersed in a fluid medium provided therein at engine operating conditions. In particular embodiments, one or more movable bodies may translate, reciprocate, oscillate, and/or

otherwise move within their corresponding chamber(s) or enclosure(s) provided within a valve. By way of example and not limitation, motion of the one or more movable bodies within a hollow chamber or enclosure, such as immersed in a fluid heat transfer medium provided therein, may be based on a motion of the operating valve, such as due to opening and/or closing motion of the valve.

[0038] FIG. 3 illustrates a cross-sectional view of an exemplary valve **105**, according to particular embodiments. In particular embodiments, valve **105** may be designed for enhanced cooling and/or for reducing valve temperatures during operation. In particular embodiments, valve **105** may be identical, equivalent, and/or otherwise usable as an exhaust valve **104**, and/or an intake valve **102**, and/or any other kind of suitable valve, and the disclosure may accordingly apply to these and any other suitable valves. Not all features illustrated by way of non-limiting example in FIG. 3, or any other figure, are required in every embodiment of use of the disclosed features. By way of example and not limitation, a valve **105** may comprise a construction without using a plug **308**.

[0039] In particular embodiments, valve **105** may be any suitable size, height, shape, and any combinations thereof. Further, valve **105** may comprise any suitable materials, such as metals, nonmetals, polymers, composites, and/or any combinations thereof. In particular embodiments, valve **105** may comprise a stem **300**, a valve head **302**, and/or a valve end **303**. In particular embodiments, valve **105** may comprise an internal chamber **304**. In particular embodiments, the stem **300** and/or valve end **303** may be configured for coupling and/or engaging with a valve actuator (such as valve actuator **202** illustrated in FIGS. 2A-2B, by way of non-limiting example). In particular embodiments, the valve head **302** may be coupled to the stem **300** at an end opposite from valve end **303**. In particular embodiments, the valve head **302** may be securely coupled to, and/or integrally formed with, stem **300**. In particular embodiments, the stem **300** may extend from the valve head **302**. In particular embodiments, a fillet **306** may be formed by the joint manufacture of the stem **300** and valve head **302**, wherein the fillet **306** may be disposed between the stem **300** and valve head **302**. In particular embodiments, during engine and valve operation, the stem **300** may transfer force and motion provided by the valve actuator to the valve head **302**. In particular embodiments, the stem **300** may support or sustain significant gas pressures received by valve head **302**. In particular embodiments, the valve head **302** may be configured to seat against an opening of the cylinder **100**, such as an opening of intake port **200** or exhaust port **204**, wherein the opening may provide fluid communication into and out of the cylinder (referring to FIGS. 1 and 2A-2B). In particular embodiments, actuation of the valve **105** such that valve head **302** may cover or uncover a particular fluid port (such as intake port **200** or exhaust port **204**) may selectively permit or deny fluid communication to the cylinder **100**.

[0040] In particular embodiments, as illustrated by way of non-limiting example in FIG. 3, the internal chamber **304** may be disposed within the stem **300**. In particular embodiments, internal chamber **304** may be sealed within the valve **105**. In particular embodiments, internal chamber **304** may further extend and terminate through a portion of the valve head **302**. In particular embodiments, internal chamber **304** may be any suitable size, height, shape, and any combinations thereof. By way of example and not limitation, as illustrated in FIG. 3, the internal chamber **304** may generally be cylindrical in shape and/or concentric or coaxial with stem **300**, but the internal chamber **304** is not limited to this shape and configuration. In particular embodiments, to form the internal chamber **304**, tube-to-solid (“TTS”) and/or top-of-head (“TOH”) manufacturing processes may be used with the stem **300** and valve head **302**. By way of example and not limitation, during TTS, the stem **300** and valve head **302** may be forged as a singular piece. In particular embodiments, a TTS operation may involve drilling through the stem **300**. By way of example and not limitation, drilling through stem **300** may comprise starting from valve end **303** opposite the valve head **302**, and continuing through the stem **300**, so as to terminate at least partially through the valve head **302**. In particular embodiments, a secondary structure may then be welded to the open end of the stem **300** to seal the formed internal chamber **304**. In particular embodiments,

during an exemplary TOH operation, the stem **300** and valve head **302** may be forged as a singular piece. In particular embodiments, the operation may then comprise drilling through the valve head **302** and continuing through the stem **300** to terminate at some location within the stem **300**. In particular embodiments, a plug **308** may then be inserted into the opening of the valve head **302** to seal the internal chamber **304**, and the plug **308** may be welded within the valve head **302**.

[0041] In particular embodiments, internal chamber **304** may be a cavity configured to receive, enclose, and/or contain a filler medium. By way of example and not limitation, a filler medium may comprise a coolant and/or a heat transfer medium. In particular embodiments, a coolant or a heat transfer medium may comprise any suitable material operable to regulate a temperature of the valve **105**, and/or to improve heat transfer characteristics of valve **105**. By way of example and not limitation, a coolant or filler medium may comprise sodium, cesium, potassium, water, and/or the like. In particular embodiments, during manufacture and assembly of valve **105**, the filler medium, which may comprise a heat transfer medium or coolant, may be inserted into the internal chamber **304** prior to sealing the internal chamber **304**.

[0042] In particular embodiments, valve **105** may further comprise one or more bodies, such as turbulator **310** by way of example and not limitation, which may be provided within internal chamber **304**. In particular embodiments, one or more bodies provided within internal chamber **304** may be configured to move within internal chamber **304**, and/or relative to the walls of internal chamber **304**, and/or relative to fixed aspects of valve **105**. While further description of one or more such movable bodies will be provided with reference to turbulator **310** to provide a non-limiting illustrative example for understanding, it should be appreciated that any suitable body or bodies provided within valve **105** are fully contemplated herein. Any disclosure provided with reference to turbulator **310** may accordingly apply to one or more movable bodies provided within valve **105**, such as within internal chamber **304**. By way of example and not limitation, the term “turbulator” used for denoting non-limiting examples of movable bodies provided within valve **105**, such as turbulator **310**, should not be considered as limiting therefore in terms of any possible or suggestible implications, such as turbulence.

[0043] In particular embodiments, turbulator **310** may be disposed within the internal chamber **304**. In particular embodiments, turbulator **310** may be at least partially disposed within a filler medium, and/or a heat transfer medium or coolant. In particular embodiments comprising a fluid phase heat transfer medium or coolant within internal chamber **304**, turbulator **310** may be at least partially immersed within a filler medium, and/or a heat transfer medium or coolant during valve operation.

[0044] In particular embodiments, turbulator **310** may be configured to facilitate mixing and/or enhanced thermal transport and/or enhanced momentum transport flow of within valve **105**. By way of example and not limitation, turbulator **310** may be configured to promote mixing and/or convection of a fluid-phase heat transfer medium or coolant provided within internal chamber **304**, such as during engine and valve operation of valve **105**. In particular embodiments, turbulator **310** may be configured to translate, reciprocate, oscillate, and/or otherwise move within the internal chamber **304** and/or interact with the heat transfer medium or coolant.

[0045] In particular embodiments, turbulator **310** may be configured to move within internal chamber **304** based on a motion of valve **105** during operation of valve **105**. By way of example and not limitation, a periodic translation and/or a reciprocal motion of valve **105** can facilitate a motion of turbulator **310** within internal chamber **304** by a transfer of energy. In particular embodiments, turbulator **310** may be configured to move within internal chamber **304** based on additional or alternative drivers of motion than a motion of valve **105**. By way of example and not limitation, a motion of turbulator **310** may be facilitated by natural convection currents in a fluid-phase coolant or heat transfer medium provided within internal chamber **304**. By way of example and not limitation, a motion of turbulator **310** may be facilitated by other forces, such as other mechanical and/or electromagnetic forces. By way of example and not limitation, electromagnetic forces, such as from a coil or driver, may act on a responsive material and/or component of

turbolator **310** to facilitate a motion of turbolator **310** within internal chamber **304**.

[0046] In particular embodiments, a movement of turbolator **310** may provide increased fluid motion of the heat transfer medium or coolant, thereby increasing heat transfer in valve **105** away from valve head **302**, such as by thermal convection, and/or such as toward valve end **303** and/or lower temperature boundary conditions for valve **105** than valve head **302**. In particular embodiments, separately or additionally, a movement of turbolator **310** may promote or increase a turbulence level of a fluid-phase heat transfer medium or coolant within internal chamber **304**, which can increase heat transfer in valve **105** away from valve head **302**, such as by thermal convection. Accordingly, a motion of turbolator **310** within valve **105** (e.g., within internal chamber **304**) can provide significantly enhanced cooling of valve **105** in operation, accordingly facilitating lower operating temperatures of valve **105**, especially but not limited to portions of valve **105** at and/or proximal to valve head **302**.

[0047] In particular embodiments, turbolator **310** may provide for increased cooling of the exhaust valve by facilitating and increase in heat transfer from the valve head **302** up through the stem **300**. In particular embodiments, certain valves **105**, such as particular exhaust valves **104**, may only transfer heat through conduction from the valve heads **302** up through the respective stems **300**. In particular embodiments, the present disclosure provides for heat transfer through both conduction and convection. In particular embodiments, turbolator **310** may provide for further increased performance through enhanced convection, such as by increasing the fluid motion and/or mixing of the heat transfer medium or coolant within the internal chamber **304**. In particular embodiments, an operating temperature reduction of valve **105** may improve fatigue life of the valve **105**, and/or may provide for usage of lower grade alloys (from high-content nickel alloy to austenitic alloy). In particular embodiments, an operating temperature reduction of valve **105** may permit use of different and/or less expensive materials and/or processes for manufacturing valve **105**. In particular embodiments, an operating temperature reduction of valve **105** may permit use of thermodynamically more efficient engine designs, cycles, or parameters, such as by permitting a higher compression ratio, and/or higher brake mean effective pressures, and/or higher levels of turbocharging than would be survivable by an engine valve not provided with one or more of the features disclosed herein.

[0048] In particular embodiments, turbolator **310** may be any suitable size, height, shape, and any combinations thereof. Further, the turbolator **310** may comprise any suitable materials, such as one or more metals. By way of example and not limitation, such as illustrated in FIG. 3, the turbolator **310** may generally be cylindrical in shape, such as a rod, but the turbolator **310** is not limited to this shape and configuration. By way of example and not limitation, turbolator **310** may generally be in the shape of a sphere, a hemisphere, a ball, a cube, a polygon, or any other suitable shape. By way of example and not limitation, turbolator **310** may comprise one or more portions that are spherical or hemispherical.

[0049] In particular embodiments, turbolator **310** may comprise an elongated portion. By way of example and not limitation, such as illustrated in FIG. 3, an elongated portion of a turbolator **310** may be oriented along a longitudinal axis of valve **105**. In particular embodiments, an elongated portion of a turbolator **310** may be provided coaxially provided with an elongated portion of internal chamber **304**. In particular embodiments, such as illustrated in FIG. 3 by way of example and not limitation, a cross-sectional dimension of turbolator **310** can be narrower than a corresponding cross-sectional inner dimension of internal chamber **304**, and/or in some non-limiting case, significantly narrower. By way of example and not limitation, a cross-sectional diameter **312** of internal chamber **304** may be between one and five times the value of a cross-sectional diameter **314** of a turbolator **310**. In particular embodiments, cross-sectional diameter **312** of internal chamber **304** may be one to ten times larger than cross-sectional diameter **314** of turbolator **310**. By way of example and not limitation, herein “diameter” may refer to a circular cross-sectional shape of one or both of internal chamber **304** and/or turbolator **310**, and/or

“diameter” may refer to another suitable length scale, such as a hydraulic diameter, of a non-circular cross-sectional shape of one or both of internal chamber **304** and/or turbolator **310**. Accordingly, in particular embodiments, turbolator **310** may exhibit rolling, yawing, tumbling, and/or other non-axial components of motion during an overall motion of turbolator **310** within **304**. In particular embodiments, a longitudinal axis of turbolator **310** may be only occasionally aligned with a corresponding longitudinal axis of internal chamber **304** and/or of valve **105** in time, or their respective longitudinal axes may not align at all.

[0050] In particular embodiments, such as illustrated in FIG. 3 by way of non-limiting example, a length of internal chamber **304** can be longer than a length of turbolator **310**. By way of example and not limitation, a length **316** of internal chamber **304** may have a value between one time and fifty times a length **318** of turbolator **310**. By way of example and not limitation, a length **316** of internal chamber **304** may have a value between one time and one hundred times a length **318** of turbolator **310**. In particular embodiments, a length of internal chamber **304** and/or turbolator **310** may be substantially aligned with a longitudinal axis of the valve.

[0051] In particular embodiments, the turbolator **310** may require, or benefit from, particular masses and/or mass distributions for accomplishing a desired movement within the internal chamber **304**. By way of example and not limitation, one or more parameters of turbolator **310** may be determined based on a balance of multiple interacting factors. By way of example and not limitation, turbolator **310** may comprise a thermally conductive material, such as one or more metals. By way of example and not limitation, some factors may comprise buoyancy forces, inertial forces, form and/or friction drag forces, gravitational forces, and/or resonance factors. In particular embodiments, turbolator **310** may comprise a shape and/or size less than the lateral dimensions of the internal chamber **304** (for example, a diameter less than the diameter of the internal chamber **304**), such as to provide for fluid flow around the turbolator **310** within the internal chamber **304** (i.e., the turbolator may not be a choking point in such examples). In particular embodiments, there may be a plurality of turbolators **310** disposed within the internal chamber **304**. In particular embodiments, a plurality of turbolators **310** may collectively function as effectively or more effectively than a singular turbolator **310**. In particular embodiments, turbolators **310** of a plural set may be identical or non-identical. In particular embodiments, a plural set of turbolators **310** may comprise turbolators laterally aligned within chamber **304**, or longitudinally aligned within chamber **304**, or any combination thereof. In particular embodiments, during manufacture and assembly of the exhaust valve **104**, the turbolator **310** or plurality of turbolators **310** may be inserted into the internal chamber **304** prior to sealing the internal chamber **304**.

[0052] FIG. 4 illustrates exemplary shapes of turbolators **310a** through **310h**, according to particular embodiments. By way of example and not limitation, such as illustrated by **310a-310d**, turbolator **310** may be asymmetric fore and aft, i.e., along a longitudinal axis of turbolator **310** and/or along a general direction of intended motion of turbolator **310** during valve operation. By way of example and not limitation, a portion of turbolator **310** may comprise a streamlined shape. By way of example and not limitation, such as illustrated by upper portions of **310a** and **310b**, a portion of turbolator **310** may comprise a tapering or narrowing section. By way of example and not limitation, such as illustrated by lower portions of **310a** or **310c**, a portion of turbolator **310** may comprise a blunt or bluff portion. In particular embodiments, an asymmetric shape of turbolator **310** may be oriented with a relatively streamlined portion pointing generally upward (i.e., generally against a gravitational direction) or pointing generally downward (i.e., generally toward a gravitational direction). In particular embodiments, such as illustrated by **310e-310h**, turbolator **310** may be symmetric fore and aft, i.e., along a longitudinal axis of turbolator **310** and/or along a general direction of intended motion of turbolator **310** during valve operation.

[0053] FIG. 5 illustrates a graph comparing operating temperature profiles of exemplary valves, according to particular embodiments. By way of example and not limitation, a temperature profile comparison is provided among otherwise comparable solid valves (labeled “solid”), sodium-filled

hollow valves (labeled “hollow”), and sodium-filled hollow valves including a turbolator **310** (labeled “enhanced HV”). A longitudinal sectional profile **510** of a valve is provided to indicate a longitudinal position corresponding to temperatures for each exemplary valve's profile. For example, position origin 0.0 indicates a cylinder-facing reference position of valve head **302** for each profiled valve, with increasing distances indicating farther longitudinal positions along stem **300** for each profiled valve. By way of example and not limitation, such as illustrated for time-averaged longitudinal profiles of operating temperature, solid valves ‘A’ and ‘B’ generally have the highest operating temperatures, followed by sodium-filled hollow valves (without turbolator **310**) ‘C’ and ‘D’. As illustrated, valves ‘E’ and ‘F’, marked “Enhanced HV” and corresponding to sodium-filled hollow valves (HV indicating hollow valve) that comprise turbolators **310**, exhibit the lowest temperature profiles during engine and valve operation. In particular, ‘E’ and ‘F’ have significantly lower temperatures at low values of longitudinal position, i.e., in and near valve head **302** which faces large heat loads and can be susceptible to failure. ‘E’ and ‘F’ also show generally lower temperatures overall, i.e., over the profile. Accordingly, the thermal performance characteristics of the presently disclosed exemplary valves **105** show lower operating temperatures than valves without one or more features disclosed herein, which lower temperatures of presently disclosed exemplary valve **105** initiate at a closer distances to the valve head **302** than do the other valves.

[0054] It will be appreciated that FIG. 5 is disclosed herein to provide examples for a better understanding, and is not to be construed as qualitatively or quantitatively limiting in any way.

[0055] Therefore, the disclosed systems and methods are well adapted to attain the ends and advantages mentioned as well as those that are inherent therein. The particular embodiments disclosed above are illustrative only, as the teachings of the present disclosure may be modified and practiced in different but equivalent manners apparent to those skilled in the art having the benefit of the teachings herein. Furthermore, no limitations are intended to the details of construction or design herein shown, other than as described in the claims below. It is therefore evident that the particular illustrative embodiments disclosed above may be altered, combined, or modified and all such variations are considered within the scope of the present disclosure. The systems and methods illustratively disclosed herein may suitably be practiced in the absence of any element that is not specifically disclosed herein and any optional element disclosed herein. While compositions and methods are described in terms of “comprising,” “containing,” or “including” various components or steps, the compositions and methods can also “consist essentially of” or “consist of” the various components and steps. All numbers and ranges disclosed above may vary by some amount.

Whenever a numerical range with a lower limit and an upper limit is disclosed, any number and any included range falling within the range is specifically disclosed. In particular, every range of values (of the form, “from about a to about b,” or, equivalently, “from approximately a to b,” or, equivalently, “from approximately a-b”) disclosed herein is to be understood to set forth every number and range encompassed within the broader range of values. Also, the terms in the claims have their plain, ordinary meaning unless otherwise explicitly and clearly defined by the patentee. Moreover, the indefinite articles “a” or “an,” as used in the claims, are defined herein to mean one or more than one of the element that it introduces.

Claims

1. A valve for a pressurizable enclosure, the valve comprising: a valve head provided at a first portion of the valve and configured to engage with the pressurizable enclosure to selectively permit fluid flow into or out of the pressurizable enclosure; a connecting portion extending between the first portion and a second portion of the valve provided opposite the first portion; a chamber comprising a hollow portion provided at least partially within the connecting portion and configured to contain a filler material; and a first body disposed within the chamber and comprising

a first portion, a second portion provided opposite the first portion, and a third portion disposed between the first portion and the second portion; wherein, during valve operation, the first body is configured to be movable within the chamber to facilitate reducing an operating temperature of the valve, and wherein one or more of the first portion of the first body or the second portion of the first body are configured to taper along a direction away from the third portion of the first body.

2. The valve of claim 1, wherein the filler material comprises a fluid medium during valve operation, and wherein the first body is at least partially immersed in the fluid medium during valve operation.
3. The valve of claim 1, wherein valve operation comprises a periodic translation or a reciprocal motion of the valve relative to the pressurizable enclosure along a longitudinal axis.
4. The valve of claim 3, wherein a motion of the first body is facilitated by the periodic translation or the reciprocal motion of the valve during valve operation.
5. The valve of claim 1, wherein the chamber comprises an elongated portion oriented along a longitudinal axis of the valve, and wherein the first body is configured to reciprocate or translate within the elongated portion of the chamber during valve operation.
6. The valve of claim 5, wherein the first body comprises an elongated portion provided coaxially with the elongated portion of the chamber.
7. The valve of claim 6, wherein a ratio of a cross-sectional diameter of the chamber and a cross-sectional diameter of the first body has a value between one and five.
8. The valve of claim 1, further comprising one or more second bodies disposed within the chamber, each second body configured to be movable within the chamber during valve operation.
9. The valve of claim 1, wherein the third portion comprises an elongated section oriented along a longitudinal axis of the valve.
10. The valve of claim 1, wherein the valve is an intake valve or an exhaust valve of a combustion engine, and wherein a motion of the first body within the chamber during valve operation permits an increased rate of heat transfer through the connecting portion of the valve to facilitate reducing the operating temperature of the valve.
11. The valve of claim 1, wherein the filler material comprises sodium.
12. The valve of claim 1, wherein reducing an operating temperature of the valve comprises reducing a mean temperature of the valve head during valve operation.
13. A method to facilitate reducing an operating temperature of a valve of a pressurizable enclosure, the method comprising: providing a movable first body within a chamber of the valve, the first body comprising a first portion, a second portion provided opposite the first portion, and a third portion disposed between the first portion and the second portion, the chamber provided at least partially within a connecting portion extending between a valve head and a valve end of the valve, the chamber configured to contain a filler material, the filler material comprising a fluid medium during valve operation, the first body at least partially immersed in the fluid medium during valve operation; and operating the valve by periodically translating the valve relative to the pressurizable enclosure along a longitudinal axis such that the first body moves within the chamber to facilitate reducing an operating temperature of the valve, wherein one or more of the first portion or the second portion of the first body are configured to taper along a direction away from the third portion.
14. The method of claim 13, further comprising: providing one or more movable second bodies within the chamber, one or more of the second bodies at least partially immersed in the fluid medium during valve operation; and operating the valve by periodically translating the valve relative to the pressurizable enclosure along a longitudinal axis such that one or more of the second bodies move within the chamber to facilitate reducing an operating temperature of the valve.
15. A cylinder assembly for an engine comprising: a first pressurizable enclosure; a first piston configured to reciprocate within the first pressurizable enclosure; and one or more first valves configured to periodically operate based on a position of the first piston to selectively permit fluid

flow into or out of the first pressurizable enclosure, wherein each first valve respectively comprises: a valve head provided at a first portion of the first valve; a connecting portion extending between the first portion and a second portion of the first valve provided opposite the first portion; a chamber comprising a hollow portion provided at least partially within the connecting portion and configured to contain a filler material; and one or more movable bodies disposed within the chamber; wherein at least one movable body of the one or more movable bodies comprises a first portion, a second portion provided opposite the first portion, and a third portion disposed between the first portion and the second portion, wherein one or more of the first portion or the second portion of the at least one movable body are configured to taper along a direction away from the third portion of the at least one movable body, and wherein, during operation of the first valve, one or more of the movable bodies are configured to translate or reciprocate within the chamber to facilitate reducing an operating temperature of the first valve.

16. The cylinder assembly of claim 15, wherein the first pressurizable enclosure comprises a cylinder of a combustion engine, wherein at least one of the one or more first valves is an inlet valve or an exhaust valve, and wherein the filler material comprises sodium.

17. The cylinder assembly of claim 15, wherein each chamber corresponding to the one or more first valves respectively comprises an elongated portion oriented along a longitudinal axis, wherein the at least one movable body corresponding to each first valve comprises a respective elongated portion, and wherein the elongated portion of the at least one movable body is provided coaxially with the elongated portion of the respective chamber.

18. The cylinder assembly of claim 15, wherein the filler material comprises a fluid medium during valve operation, and wherein one or more of the movable bodies are at least partially immersed in the fluid medium during valve operation.

19. The cylinder assembly of claim 15, wherein, for each first valve, periodic operation comprises a periodic translation or a reciprocal motion of the first valve relative to the first pressurizable enclosure.

20. The cylinder assembly of claim 15, further comprising: one or more second pressurizable enclosures; one or more second pistons respectively configured to reciprocate within the one or more second pressurizable enclosures; and one or more second valves corresponding to each of the one or more second pressurizable enclosures, each second valve respectively configured to periodically operate based on a position of the corresponding second piston to selectively permit fluid flow into or out of the corresponding second pressurizable enclosure, wherein each second valve respectively comprises: a valve head provided at a first portion of the second valve; a connecting portion extending between the first portion of the second valve and a second portion of the second valve provided opposite the first portion of the second valve; a chamber comprising a hollow portion provided at least partially within the connecting portion of the second valve and configured to contain a filler material; and one or more movable bodies disposed within the chamber of the second valve; wherein, during operation of the second valve, one or more of the movable bodies are configured to translate or reciprocate within the chamber of the second valve to facilitate reducing an operating temperature of the second valve.
